Republic of the Philippines
Department of Science and Technology
PHILIPPINE COUNCIL FOR AGRICULTURE, AQUATIC AND NATURAL RESOURCES RESEARCH AND DEVELOPMENT
Los, Baños, Laguna

MEMORANDUM OF AGREEMENT

On

SMS in Aquafeeds: Development of an Alternative Aquafeed Ingredient Using Spent Mushroom Substrate (SMS)

KNOWN ALL MEN BY THESE PRESENTS:

This agreement entered into by and between:

The **DEPARTMENT OF SCIENCE AND TECHNOLOGY - PHILIPPINE COUNCIL FOR AGRICULTURE, AQUATIC AND NATURAL RESOURCES RESEARCH AND DEVELOPMENT** hereinafter referred to as **"DOST-PCAARRD"**, with principal office at Los Baños, Laguna and represented by its Executive Director, **DR. REYNALDO V. EBORA**;

-and-

The **PARTIDO STATE UNIVERSITY**, represented herein by its President, **DR. RAUL G. BRADECINA**, herein referred to as the "**ParSU**", for the implementation of the above-mentioned project;

WHEREAS, the abovementioned Project was approved by the **PCAARRD Directors' Council** on May 16, 2023 via Zoom Conference.

NOW THEREFORE, for and in consideration of the foregoing premises, the PARTIES hereby agree to the following terms and conditions:

WITNESSETH:

I. Purpose

This agreement seeks to implement the project titled **"SMS in Aquafeeds: Development of an Alternative Aquafeed Ingredient Using Spent Mushroom Substrate (SMS)"**.

II. Objectives

General Objective

To formulate and produce an effective aquafeed in tilapia aquaculture using locally derived ingredients, the spent mushroom substrate (SMS).

Specific Objectives

1. To characterize the nutritional value of spent mushroom substrate;
2. To formulate and develop aquafeed using SMS as an alternative ingredient;
3. To assess the nutritional and functional properties of the developed aquafeeds;
4. To conduct feeding trial and assess growth performance in tilapia;
5. To determine cost and return analysis of the developed aquafeeds; and
6. To assess dietary intake of aquafeeds using metagenomics studies.

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Supervising Administrative Officer
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Project Leader, ParSU

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III. Expected outputs

6Ps	Year 1	Year 2
Publication	None	At least 1 draft publication submitted in ISI/Scopus Journal; At least 1 paper presentation in conference; and At least 1 IEC material produced
Patent	None	At least 1 IP protection for the SMS Aquafeed
Product	None	One (1) product (SMS aquafeed)
People Services	Mentored/ Trained at least 2 researchers	Mentored/ Trained at least 2 researchers and 2 undergraduate students; Trained at least 2 local farmers in the production of SMS aquafeeds.
Places and Partnership	MOA with UPLB, DA RFO5 and Bicol Entrepreneurship Association for Mushroom (BEAM, Inc.)	
Policy		One (1) policy recommendation for potential use of SMS aquafeed in aquaculture farming.

2Is	
Social Impact	This project will help promote awareness to farmers and various stakeholders in using alternative ingredients such as SMS in formulating aquafeeds for tilapia. The use of SMS as an alternative aquafeed ingredient will offer alternatives in achieving higher-quality aquafeed. This will also help reduce the tons of SMS produced in the mushroom industry by utilizing its generated wastes.
Economic Impact	The use of SMS as an alternative ingredient for aquafeed formulation can help reduce the feed costs and improve profit margins. This also promotes sustainable practices by using waste products generated by the mushroom industry.

IV. Duration of the Project

The project shall be implemented for 2 years, covering the period November 1, 2023 – October 31, 2025. The project shall commence immediately or within two (2) months after the release of funds. In case there is a change in actual date of implementation, it shall be requested from **DOST-PCAARRD** by **ParSU**.

V. Rights and Obligations of the Parties

DOST-PCAARRD shall:

a. Provide financial support for the project with a total grant **FOUR MILLION NINE HUNDRED FIFTY-THREE THOUSAND EIGHT HUNDRED NINETY-SIX AND 70/100 PESOS (PhP 4,953,896.70)** with the following breakdown:

Agency	Year 1	Year 2	Total
ParSU	2,447,018.20	2,506,878.50	4,953,896.70

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For its Y1 implementation, the total amount of **TWO MILLION FOUR HUNDRED FORTY-SEVEN THOUSAND EIGHTEEN AND 20/100 PESOS (PhP 2,447,018.20)** shall be remitted to implementing agency, the line-item-budget of which is attached as **Annex A** and made an integral part hereof.

Such grant shall be for the Year 1 of the project to be led by **ParSU** and to be expended as allocated in the project's approved line-item-budget and released subject to terms and conditions specified in the "Guidelines for the Grants-In-Aid Program of Department of Science and Technology and its Agencies (DOST AO No. 011, series of 2020), effective September 2020 and PCAARRD Supplemental Provisions (PCAARRD AO No. 198.B, series of 2019), effective August 02, 2019 (see **Annex B**) shall govern the implementation of the project.

Release of succeeding project funds shall be subject to review, evaluation and approval by **DOST-PCAARRD** approving bodies duly supported by the required financial and technical reports.

DOST-PCAARRD shall deduct the project's unexpended balance and savings of prior project year from the funds of the succeeding project year, e.g., unexpended balance in Year 1 shall be deducted from Year 2 project funds to be released, unless the use is approved by **DOST-PCAARRD** approving bodies.

b. Monitor and evaluate the implementation of the project in accordance with the objectives and the workplan;

c. Conduct periodic field evaluation of the project to identify problems, solutions and remedial actions to avoid delays in implementation;

d. Provide adequate coordinative and efficient administrative support and management system necessary for the smooth implementation of the program/projects;

e. Ensure that specific work plans agreed by all parties are adhered to including budgets necessary to implement such plans;

f. Review periodic financial reports submitted by the implementing agency; and

g. Appraise technical reports (semi-annual, annual, publishable report and terminal).

The implementing agency, ParSU through their Head, **DR. RAUL G. BRADECINA**, bind themselves to implement the project under an assigned Project Leader in **ParSU** and in accordance with the approved project proposal (made part hereof as **Annex C**) and shall be strictly implemented following the project's work and financial plan.

Among the responsibilities of the implementing agency as provided for in these guidelines are as follows:

ParSU shall:

1. Exert all efforts (such as providing facilities and technical personnel with the required expertise) to attain the objectives stated in the proposal; and

2. Take the lead in the implementation of project activities and coordinate with other collaborating agencies;

3. Hire the required project personnel as indicated in the approved proposal in accordance with the existing hiring policies of the agency and the approved line-item budget; request approval from **DOST-PCAARRD** of any changes in the position/designation of the project personnel to be hired; and submit list of personnel engaged in the project following DOST Form 11.

4. Submit the following technical accomplishment reports duly endorsed by the Agency Head to **DOST-PCAARRD**:

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4.1 An acceptable semi-annual report using DOST Form 6 in two (2) hard copies and one (1) electronic copy within a month after the end of the project semester.

4.2. An acceptable annual technical accomplishment report using DOST Form 7 in two (2) hard copies and one (1) electronic copy within two (2) months after the end of a project year. In case of a program, a consolidated annual program report shall be submitted in addition to the individual project reports.

4.3. A quality terminal technical report following DOST Form 15 in three (3) hardcopies and one (1) electronic copy within three (3) months after project completion which shall be accompanied by two (2) hardcopies and one (1) electronic copy of publishable technical report. The publication of such paper in any existing scientific or professional journal should meet the standards set by said journal.

5. Disburse funds in accordance with the approved Line-Item Budget (LIB), subject to existing government accounting and auditing procedures, and make necessary reprogramming of funds for not more than three (3) times in each implementation year, following these rules:

5.1 Approve, through Implementing Agency Head or his authorized representative, any reprogramming/transfer of funds of existing expense items (other than those enumerated in 5.2) budget as originally approved by the **DOST-PCAARRD** to augment direct and indirect cost under PS and MOOE. A copy of approval shall be provided to **DOST-PCAARRD** within two (2) weeks for documentation/concurrence of the monitoring division, otherwise the approved reprogramming shall be null and void;

5.2. Exceptions to the provisions of 5.1 are the following:

5.2.1 Budget for the Foreign Travel and Training

5.2.2 Creation of new expense item, including increase/decrease of numbers of items (e.g., equipment, personnel) originally approved in the LIB and upgrading/downgrading of positions from the originally approved PS

5.2.3 Increase in indirect cost budget exceeding the 7.5% ceiling for indirect cost

6. The use of foreign travel funds, if provided for in the approved LIB, shall be governed by DOST Administrative Order No. 001, series of 2017: Revised Rules on Foreign Travel of DOST Personnel and Rule Governing the Use of Travel Grants for Paper/Poster Presentation, effective 29 January 2018.

7. Submit at least three (3) copies of the following financial reports to DOST-PCAARRD:

7.1. Semi-annual financial report using DOST Form 8, certified correct by the agency accountant and approved by the agency head within a month after each semester, supported by Report of Disbursement (ROD).

7.2. Annual financial report using DOST Form 8 certified correct by the Agency Accountant and approved by the agency head within two (2) months after end of each implementation year.

7.3. Terminal audited financial report (TAFR) using DOST Form 16 within three (3) months after project completion. The reported unexpended balance shall be refunded back to **DOST-PCAARRD**.

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For private higher education institutions (HEIs) and research and development institutions (RDIs), an audited FR shall be submitted, duly certified by a licensed accountant together with an Audit Certificate/Report.

Financial reports are accompanied by DOST Forms 9, 11, 12, and 13 whenever applicable.

8. Be accountable for equipment purchased for the project. Only those equipment listed in the approved proposal shall be purchased by the agency following the government Procurement Reform Act. **DOST-PCAARRD** remains to be the owner of all equipment purchased out of its grant and shall have the right to transfer/assign/donate such to anyone after the project completion.
   Enter all equipment purchased in the institution's books on inventory of equipment. Submit report on equipment purchased using DOST Form 12 with the corresponding Property Acknowledgement Receipts (PARs).

9. Make available to authorized officials of **DOST-PCAARRD** and the Commission on Audit during office hours, the records of project operations and books of accounts for inspection whenever necessary.

10. Present the project results in agency in-house reviews/program review and/or at least in one seminar or scientific meeting for the purpose of disseminating the information and technology generated from the project; without prejudice to the provision of Section VI hereof.

Program/Project Leader, shall:

a. Coordinate with Program/Project Leader to ensure that goals of the project and program are attained;

b. Facilitate and monitor the implementation of project activities;

c. Coordinate, facilitate and monitor the preparation of the project's accomplishments and financial reports which shall be packaged by the Program Leader as a program output;

d. Consolidate project's accomplishment reports and prepare syntheses of accomplishment reports; and

e. Maintain close communication and coordination with **DOST-PCAARRD**.

Project Staff, shall:

a. Coordinate with Project Leader to ensure that goals of the project are attained;

b. Facilitate and monitor the implementation of project activities;

c. Facilitate the preparation and submission of the project's accomplishments and financial reports which shall be packaged by the Project Leader as a project output; and

d. Maintain close communication and coordination with **DOST-PCAARRD**.

VI. Ownership and Utilization of Information, Technologies, Inventions and Improvements

The **Implementing Agency** and the researchers recognize that results of the programs/projects and the rights to publication, except those that relate to intellectual property concerns, belong jointly to the researchers and implementing

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agencies. Ownership of intellectual property and intellectual property rights shall be governed by but not limited to, Chapter II, Rule 5 and Rule 6 of the Implementing Rules and Regulations of Republic Act (R.A.) 10055.

Any publication arising from the research and other activities undertaken by virtue of this agreement shall identify the Implementing Agency/ies as the source of the output.

Full acknowledgments shall accordingly be given to **DOST-PCAARRD** in case the research results are published or presented in various forums, seminars, meetings. Such presentation should not prejudice the proprietary/confidential nature of the information as heretofore mentioned for purposes of intellectual property protection.

Subject to the limitations recognized by law, particularly those that relate to intellectual property, the implementing agencies reserve the right to use all data and findings by virtue of and pursuant to this agreement for the enhancement of its academic and research programs.

VII. Discontinuance of Assistance

DOST-PCAARRD reserves the right to:

a. Discontinue any program/project of its assistance at anytime with prior consultation with Implementing Agency for violation of this agreement or when it is determined that the results obtained or are reasonably expected to be obtained do not justify further activity for reasons attributable to the willful act of the project leader and/or implementing team; Suspend the program/project and the disbursement of funds for violation of any part of this agreement;

b. Terminate any project when funds are not available from the national government.

Discontinuance of assistance, suspension or termination of this agreement shall be done by **DOST-PCAARRD** upon prior consultation with the **ParSU**. In case of suspension, upon receipt of the written notice therefore, the implementing agency shall cease disbursement of project funds and shall resume only upon the written advice of **DOST-PCAARRD**. In case of termination, **DOST- PCAARRD** shall notify the Program/Project Leader at least forty-five (45) days before the date of termination so that he/she could accordingly notify the program/project staff at least a month before the date of project termination.

VIII. Amendment Provision

Any additional or amendments to the stipulations hereof shall be done through a supplementary agreement to be prepared by the party requiring such and shall be mutually consented by the parties. Notarization of such shall be taken care of by the party who prepared the amendments/additional provisions.

IX. Repealing Clause

All parties agree that this MOA contains the full agreement of the parties and supersedes all previous agreements either written or oral if there are any.

X. Alternative Dispute Resolution

All disputes, controversies, or claims arising out or relating to this contract, shall be settled through negotiation and/or mediation and shall be decided by arbitration in accordance with the rules of the UNICITRAL Law as adopted in RA 9285 (Alternative Dispute Resolution Law of 2004). This is, however, without prejudice

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to the provision of Rule 26 of RA 10055 on resolution of disputes, particularly on matters pertaining to ownership.

IN WITNESS WHEREOF, the parties hereto have hereunto set their hands, this _____ at _____.

DOST- PCAARRD:

REYNALDO V. EBORA
Executive Director

ParSU:

RAUL G. BRADECINA
President

WITNESSES:

ADELAIDA T. CALPE
Director, IARRD

LIEZEL DC. ATOLE
Project Leader, ParSU

Certified Funds Available:

ABEGAIL GRACE M. MARALIT
Supervising Administrative Officer
OIC, FAD-Accounting, PCAARRD

ACKNOWLEDGMENT

Republic of the Philippines)
Municipality of Los Baños, Laguna) S.S.

BEFORE ME, A NOTARY PUBLIC for and in Los Baños, Laguna, Philippines, on this JUL 12 2023 2023, personally appeared:

NAME	COMM. TAX RECEIPT NO. PASSPORT NO.	DATE/PLACE ISSUED
REYNALDO V. EBORA	Passport No. S0026509A	Oct. 6, 2022 / DFA Manila
RAUL G. BRADECINA		

Known to me to be the same persons who executed the foregoing instrument and they acknowledged to me that the same is their free and voluntary act and deed as well as that of the Corporations/Entities they represent.

The foregoing, which refers to a Memorandum of Agreement, consists of eight (8) pages including this page wherein this Acknowledgment is written, and has been signed by the parties and witnesses on all pages.

WITNESS MY HAND AND SEAL on the date and place above-written.

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ATTY. ELENO O. PERALTA
Notary Public Until December 31, 2023
Calamba City, Los Baños, Bay, Calauan, Laguna
IBP No. 187857/02-12-2022/Laguna Chapter
Not. Com. # 14-2022-C dated 01-06-2022
PTR No. 1058882/01-03-23/Los Baños, Laguna
MCLE Compliance No. VII-0004580/August 25, 2021
Roll No. 33880/May 1986